

XAI Lecture 24 — Explaining LLM Reasoning

Background, then two papers: *generate* the reasoning (Rajani 2019) and *localise* the evidence (Yin & Neubig 2022)

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

🍷 Background - the problem before the two papers ---

Both papers attack the same question — **how do language models (LMs) reason?** — from opposite ends. Before that, three things must be on the table:

- **What commonsense reasoning is**, and why it is hard for NLU (Natural Language Understanding) systems.
- **ConceptNet** — the knowledge graph (Speer et al., 2017).
- **CommonsenseQA (CQA)** — the question-answering (QA) benchmark built *from* ConceptNet (Talmor et al., 2019), used by both papers.

Why start here

CQA is not a generic multiple-choice set. *How its questions are manufactured* is exactly what makes them hard — and it explains a surprising result we will meet in Paper 1.

🍷 Why commonsense is hard for NLU ---

When people answer, they draw on world knowledge **not present in the text**: space, cause and effect, social conventions.

Example (Talmor et al., 2019):

“Where was Simon when he heard the lawn mower?”

A human silently infers: a lawn mower is **outdoors**, at **street level** → Simon was **outside**.

The contrast

Classic QA (e.g. SQuAD) hands you a paragraph and asks about *it*.

Commonsense QA gives **no** such paragraph — the knowledge must already be in the model.

Remarks

“Trivial for humans, out of reach for NLU systems” — the gap this whole lecture circles around.

🍎 ConceptNet: a commonsense knowledge graph (Speer et al., 2017) ---

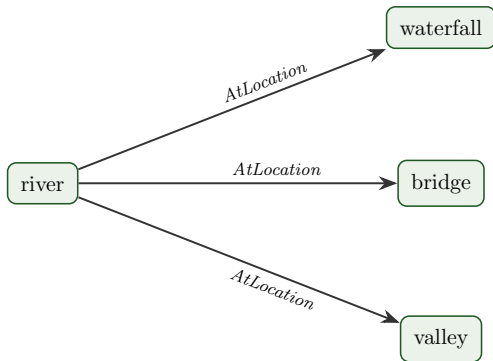
ConceptNet stores everyday knowledge as a **graph of triples**:

- nodes are **concepts** (words / phrases);
- edges are **named relations**.

Typical relations: `IsA`, `AtLocation`, `UsedFor`, `CapableOf`, `Causes`, `PartOf`, `HasProperty`.

Remarks

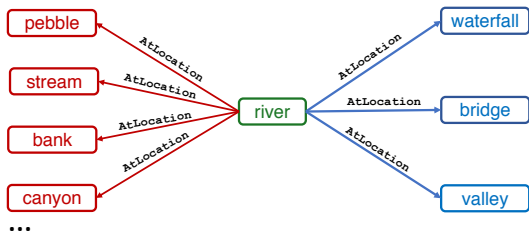
Common sense *written down* as a graph: millions of obvious facts, each a typed arrow between two ideas.



Other triples: `knife-UsedFor->cutting`,
`bird-CapableOf->fly`.

🍷 From ConceptNet to questions: how CQA is built ---

- a) Sample ConceptNet for specific subgraphs



- b) Crowd source corresponding natural language questions and two additional distractors

*Where on a **river** can you hold a cup upright to catch water on a sunny day?*

✓ **waterfall**, ✗ bridge, ✗ valley, ✗ pebble, ✗ mountain

*Where can I stand on a **river** to see water falling without getting wet?*

✗ waterfall, ✓ **bridge**, ✗ valley, ✗ stream, ✗ bottom

*I'm crossing the **river**, my feet are wet but my body is dry, where am I?*

✗ waterfall, ✗ bridge, ✓ **valley**, ✗ bank, ✗ island

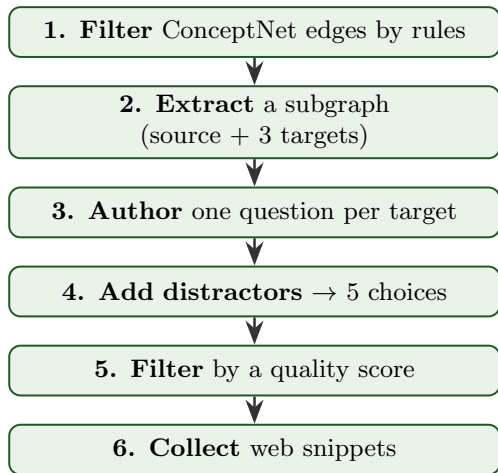
Talmor et al. (2019), Fig. 1.

A worker sees a **source concept** (*river*, green) and **three target concepts** (*waterfall*, *bridge*, *valley*, blue) sharing one relation (*AtLocation*).

The worker writes **three questions**, one per target as the answer; the other two are distractors.

Then per question: **+1 ConceptNet distractor** (red) and **+1 hand-authored** (purple) → **5 choices** total.

🍷 The generation pipeline (Talmor et al., 2019) ---



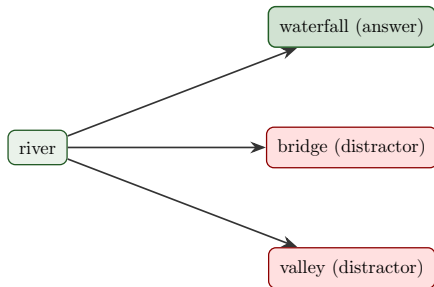
Six construction stages, top to bottom:

- ① **Filter** ConceptNet edges with rules.
- ② **Extract** a subgraph (source + 3 targets).
- ③ Workers **author** one question per target.
- ④ Workers **add distractors** (→ 5 choices).
- ⑤ Workers **filter** by a quality score.
- ⑥ **Collect web snippets** (for search baselines).

After Talmor et al. (2019).

🍷 Why the distractors make it hard ---

The distractors are **ConceptNet siblings**: same source concept, same relation.



So a model **cannot** win by spotting which option “sounds river-ish” — they all do. It must read the specific situation (*hold a cup upright to catch water*).

Key insight

The construction *forces* commonsense: surface word-association is neutralised by design, because every option is equally associated with the source.

🍷 The numbers, and the gap ---

- **12,247** questions in total (Talmor et al., 2019).
- Best baseline **BERT-large** $\approx 56\%$ vs **human** $\approx 89\%$.
- Even adding **Google-search snippets** does **not** help $\rightarrow$ the difficulty is reasoning, not missing data.

Two versions — remember this!

v1.0: $\approx 9,500$ questions, **3** choices (GPT 54.8%, human 95.3%). **v1.11**: 12,247 questions, **5** choices.

Why it returns

Paper 1 reports results on *both* versions. The jump from 3 to 5 ConceptNet-sibling choices is exactly where its method will stumble.

🍷 Background → the two papers ---

The set-up for the lecture

Given this hard benchmark, two natural questions follow:

- **Paper 1 (Rajani et al., 2019):** can a model that *writes its reasoning* answer better?
→ *generate* explanations.
- **Paper 2 (Yin & Neubig, 2022):** can we *point at the exact input word* that drove a prediction? → *localise* evidence.

verbalise the reasoning

vs.

localise the evidence

🍎 The two papers at a glance ---

Unit IV · Week 12. Both methods explain a trained **language model (LM)** *from the outside* (inputs → outputs) — but the LM itself differs per paper.

Paper 1 · Rajani et al. (2019) — CAGE

Verbalise the reasoning

- **Explains:** a BERT classifier (few answer options).
- **What:** a natural-language narrative.
- **Goal:** plausibility + accuracy.
- **Family:** generated explanation.
- **Nature:** black-box (internals not inspected).

Paper 2 · Yin & Neubig (2022)

Localise the evidence

- **Explains:** a generative LM — GPT-2 / GPT-Neo (~ 50,000 tokens).
- **What:** attributes the output to input tokens.
- **Goal:** faithfulness, fine-grained evidence.
- **Family:** contrastive attribution (gradient/erasure).
- **Nature:** gradients (white-box) or perturbation.



Explain Yourself! Leveraging Language Models for Commonsense Reasoning

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Proceedings of the 57th Annual Meeting of the ACL, 2019 (arXiv:1906.02361)

Introduction + Motivation ---

The problem: deep models do poorly on tasks needing commonsense reasoning — knowledge not present in the input.

- An **explanation** verbalises the reasoning a model uses.
- **CommonsenseQA** is the benchmark (Talmor et al., 2019) — *see Background*.
- Open question: *do* these models reason, and how much rests on world knowledge?

Core idea

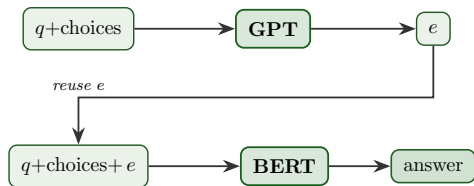
Train a language model to **generate explanations**, feed them to a classifier, and **measure whether they help**.

Remarks

Test whether the explanation *helps accuracy* — not merely whether it sounds nice.

🍷 The pipeline, end to end ---

- 1 Take a CQA example: question q , choices c_0, c_1, c_2 , gold answer a .
- 2 A human writes explanation e_h for *why* a is correct (this is **CoS-E** — Common Sense Explanations).
- 3 Fine-tune a language model (**GPT** — generative pre-training, Radford et al., 2018) to generate $e \approx e_h$.
- 4 Concatenate $q+\text{choices}+e$ and feed to a **BERT** classifier (bidirectional encoder, Devlin et al., 2019).
- 5 Measure: does e raise accuracy?



Remarks

GPT explains, BERT decides. The LM is a commentator, not the judge.

🍷 Didactic aside: what does “fine-tune GPT to generate e” mean? ---

- A language model already knows how to **continue text** (next-word prediction).
- **Fine-tuning** = keep training it, but now on (question + choices) --> explanation pairs from CoS-E.
- After fine-tuning, given a *new* question it can **write its own explanation**, imitating the human ones.

Mental model

We are not teaching new facts; we are teaching a *format*: “given a question, produce the kind of one-sentence justification a person would write.”

🍷 CoS-E: Common Sense Explanations ---

A new dataset **on top of CQA**, collected via Amazon Mechanical Turk.

- **CoS-E-selected**: highlighted words in the question.
- **CoS-E-open-ended**: a free-form explanation sentence.
- Sizes (train / dev): v1.0 = 7,610 / 950; v1.11 = 9,741 / 1,221.
- Quality control: ≥ 1 highlighted word, explanation ≥ 4 words, not a substring, templates filtered.

Remarks

Before GPT can *write* explanations, it needs *examples* of explanations. CoS-E is that supervision.

🍷 CoS-E: three concrete examples (Table 1) ---

From the paper

Q: “While eating a hamburger with friends, what are people trying to do?” Choices: **have fun**, tasty, indigestion.

CoS-E: “Usually a hamburger with friends indicates a good time.”

Q: “After getting drunk people couldn’t understand him, it was because of his what?” Choices: lower standards, **slurred speech**, falling down.

CoS-E: “People who are drunk have difficulty speaking.”

Q: “People do what during their time off from work?” Choices: **take trips**, brow shorter, become hysterical.

CoS-E: “People usually do something relaxing, such as taking trips, when they don’t need to work.”

🍷 CoS-E: a worked example, link by link ---

Question

While eating a hamburger with friends, what are people trying to do? — **have fun** / tasty / indigestion.

The bare question never says friends are enjoyable.

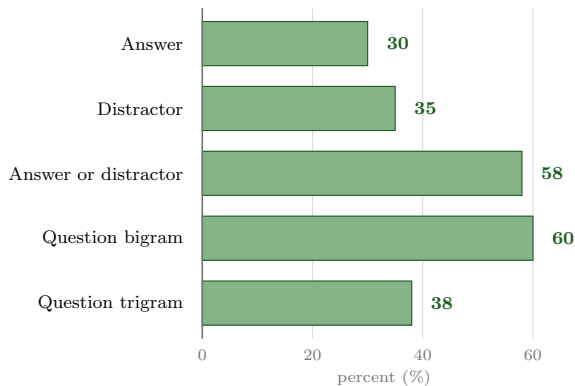
The explanation supplies the missing world-knowledge link:

hamburger + friends → social → enjoyable → **have fun**

Remarks

That chain — “hamburger with friends = a good time” — is commonsense made *explicit in words*. That is what CoS-E records.

🍷 CoS-E: what it actually contains ---



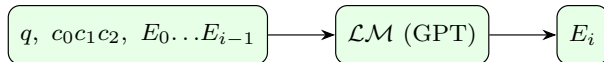
- **58%** of explanations contain the ground-truth answer.
- Even explanations with **no** word overlap with any choice still beat the no-explanation baseline.

Takeaway

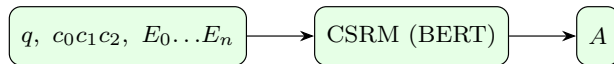
The explanation adds *real signal*, not mere repetition of the answer.

CAGE: Commonsense Auto-Generated Explanations ---

(a) generate



(b) classify



- **Phase 1 — generate:** fine-tune GPT on CQA + CoS-E to produce e .
- **Phase 2 — classify:** BERT predicts the answer using $q + \text{choices} + e$.

Recall

First *manufacture* the “why”; then *decide* with its help.

🍷 CAGE-reasoning (explain-then-predict) ---

Main approach. LM conditioned on question + choices + human explanation, **not** the gold label.

Input context (the prompt C_{RE}):

“ q , c_0 , c_1 , or c_2 ? commonsense says ___”

Objective:

$$\max_{\Theta} \sum_i \log P(e_i \mid e_{i-k}, \dots, e_{i-1}, C_{RE}; \Theta)$$

Memory hook

The prompt ends in **commonsense says ___: no answer inside**. So the explanation is produced *before* the answer is known.

🍷 Deep dive: the training objective, term by term ---

Decompose $\max_{\Theta} \sum_i \log P(e_i \mid e_{<i}, C_{RE}; \Theta)$.

- e_i — the i -th token of the explanation to produce.
- $e_{<i}$ — tokens already written (read to continue).
- C_{RE} — prompt: question + choices + “commonsense says”.
- Θ — the LM parameters being fine-tuned.
- $\sum_i \log P$ — next-token likelihood over explanation tokens.

Remarks

Ordinary next-word training, but the “text to predict” is the *explanation*, and the prompt *excludes the label* — so nothing leaks the answer.

🍷 Didactic aside: “label in the prompt” - the same example, both ways ---

Reasoning (C_{RE}) — answer absent

“While eating a hamburger with friends, what are people trying to do? **have fun**, tasty, or indigestion?
commonsense says _____”

Rationalisation (C_{RA}) — answer present

“While eating a hamburger with friends ...? have fun, tasty, or indigestion? **have fun** *because* _____”

Only one word's worth of difference — but in one case the model already knows the answer.

🍷 CAGE-rationalization (predict-then-explain) ---

The reverse. LM conditions on the **predicted label**; produces a post-hoc justification.

$C_{RA} = "q, c_0, c_1, \text{ or } c_2? \text{ a because } ____"$

Hyper-parameters: max length 20, batch 36, ≤ 10 epochs, lr $1e-6$, warmup 0.002, weight decay 0.01.

Remarks

The explanation comes *after* the answer
→ a **rationalisation**, not reasoning.

More interpretable, but it cannot be the
cause of the prediction.

🍷 Deep dive: reasoning vs. rationalisation ---

One word changes everything: where the label sits.

- **Reasoning** (C_{RE}): no label $\rightarrow$ e made at inference, used as genuine new context.
- **Rationalisation** (C_{RA}): label in prompt $\rightarrow$ e conditioned on the answer.
- Accuracy: reasoning **+10%** over the previous state of the art (SOTA); rationalisation **+6%**.
- Only reasoning can be called “commonsense reasoning”.

Key insight (the heart of the lecture)

Faithfulness is decided by **information flow**, not output quality. A *fluent* rationalisation can still be *unfaithful*.

🍷 Results: CommonsenseQA v1.0 (3 choices) ---

Dev (random split)

Method	Acc. (%)
BERT baseline	63.8
CoS-E-open-ended	65.5
CAGE-reasoning	72.6

Test split (v1.0)

Method	Acc. (%)
RC (Talmor)	47.7
GPT (Talmor)	54.8
CoS-E-open-ended	60.2
CAGE-reasoning	64.7
Human	95.3

RC = reading-comprehension baseline (Talmor et al., 2019).

Headline

≈ 10% absolute over previous SOTA on test — yet still far below the human 95.3%.

🍷 Results: CQA v1.11 (5 choices) - the uncomfortable result ---

Method	Acc. (%)
CAGE-reasoning	55.7
BERT baseline	56.7
CoS-E-open-ended	58.2

Remarks

On the harder 5-choice version **CAGE loses to the plain baseline**. The explanation often *contains* the right answer, but the classifier cannot exploit it.

Connect to Background

Recall: v1.11 has **five ConceptNet siblings**. With options that semantically close, simply *concatenating* the explanation is not enough. The authors report this honestly.

🍷 Results: out-of-domain transfer ---

Method	SWAG	Story Cloze
BERT	84.2	89.8
+ explanation transfer	83.6	89.5

Remarks

Transfer with no retraining costs only a **tiny drop** ($< 0.6\%$). Fluent, relevant explanations — but no downstream gain. *An honest negative result.*

🍷 Qualitative analysis ---

- CAGE explanations use **simpler constructions** than humans, yet can be *more* informative.
- Contain an answer choice **43%** of the time; the *predicted* choice only **21%**.
- **BLEU** (n-gram overlap with references) vs human explanations peaks at **4.1** (vs 0.8 untuned); perplexity 32.

Foreshadowing

Low BLEU but real usefulness → something can be *useful without resembling human wording*. This previews Paper 2's **faithfulness vs. plausibility** tension.

🍷 Limitations + critical reading of Rajani et al. ---

- **Human simulatability is low:** from the explanation alone, Turkers recover the model's answer **42%** (CAGE) vs **52%** (human) → the explanation does not transparently reveal the model.
- **Adversarial explanations are catastrophic:** misleading explanations drop accuracy **60%** → **30%** — below the 50% baseline.
- **Bias propagation:** CQA gender disparity flows into CoS-E and the trained models.

🍌 Bridge to Paper 2 ---

What Rajani et al. leave open

They *generate* explanations, but never ask: **which input tokens caused this prediction**, and **why this token instead of another?**

- Generated explanations are **plausible** but not necessarily **faithful**.
- Language generation has an enormous output space → we need a **token-level** lens.

Next

Yin & Neubig (2022): contrastive input-saliency — look *inside* the model.

Interpreting Language Models with Contrastive Explanations

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Proceedings of EMNLP 2022 (Outstanding Paper Award) (arXiv:2202.10419)

🍷 Introduction + Motivation: the output-space problem ---

- Classification: **small** output space (few labels).
- Language modelling: **tens of thousands** of tokens per step.
- A single prediction conflates **many** decisions: part of speech, number, tense, semantics.

Symptom

Non-contrastive saliency just highlights the token **right before** the prediction — uninformative about subtle choices.

🍷 Didactic aside: “output space”, concretely ---

- A spam classifier picks among {spam, ham} — 2 options. “Why?” is easy.
 - A language model picks the next token among $\sim 50,000$ options.
- “Can you stop the dog _____” $\rightarrow$ {barking, crying, walking, running, ...}

Why this matters

With 50,000 outputs, “what mattered for the prediction?” blurs together grammar, number, tense and meaning into one uninformative answer.

The contrastive lens ---

Why did the model predict “**barking**” given “*Can you stop the dog from ___*”?

- Non-contrastive: highlights “from” (obvious preceding token).
- “barking” **instead of** “crying”? → “**dog**” matters.
- “barking” **instead of** “walking”? → “**stop**” matters.

Key insight

Contrastive explanation (Lipton, 1990) asks why target y_t *rather than* foil y_f . The **foil** picks *which* conflated decision we explain.

🍷 Didactic aside: what is a “foil”? ---

- y_t = the **target** = what the model actually said (*barking*).
- y_f = the **foil** = a chosen rival you compare against (*crying, walking, ...*).

barking vs crying → **dog** barking vs walking →
 stop

The trick

Change the foil, change the *question*.
The foil is how you *aim* the explanation
at one specific decision.

🍷 Formal definition: contrastive saliency ---

Non-contrastive gradient:

$$g(x_i) = \nabla_{x_i} q(y_t | x)$$

Contrastive gradient:

$$g^*(x_i) = \nabla_{x_i} (q(y_t | x) - q(y_f | x))$$

Remarks

∇ just means “how much it influences”.
Not “what raised y_t ?” but “what raised y_t *while lowering* y_f ?” — the subtraction cancels shared evidence.

🍷 Deep dive: why the subtraction works ---

Walk $g^*(x_i) = \nabla_{x_i}(q(y_t | x) - q(y_f | x))$:

- 1 $\nabla_{x_i} q(y_t | x)$ — how x_i pushes the **target** up.
- 2 $\nabla_{x_i} q(y_f | x)$ — how the **same** token pushes the **foil** up.
- 3 Subtract: evidence raising *both* (“predict some verb-ing”) cancels.
- 4 What survives separates y_t from y_f specifically.
- 5 Norm $\Rightarrow$ magnitude; gradient $\times$ input $\Rightarrow$ signed direction.

Remarks

Generic syntactic pressure is shared, so it disappears. The leftover is the **semantic / long-range** cue we wanted.

🍷 Method 1: Contrastive Gradient Norm ---

$$S_{GN}^*(x_i) = \|g^*(x_i)\|_{L_1}$$

- White-box, cheap (one backward pass).
- **Magnitude** only, no direction.

Use it for

“Does this token matter at all?” — it tells you *how strongly* x_i tips the decision, not *which way*.

🍷 Method 2: Contrastive Gradient $\times$ Input ---

$$S_{GI}(x_i) = g(x_i) \cdot x_i, \quad S_{GI}^*(x_i) = g^*(x_i) \cdot x_i$$

- White-box, cheap, and **signed**.
- Multiplying by the embedding weights influence by how present the token is.

Use it for

“Towards y_t (positive) or *towards* the foil (negative)?” — more information than Method 1.

🍷 Method 3: Contrastive Input Erasure ---

$$S_E^*(x_i) = (q(y_t | x) - q(y_t | x_{\neg i})) - (q(y_f | x) - q(y_f | x_{\neg i}))$$

- **Black-box:** remove x_i and measure the change.
- **Expensive:** one forward pass per token.

Remarks

$x_{\neg i}$ is the sentence *without* that word.
Delete it: if y_t drops and the foil y_f rises, the word mattered.

🍷 Deep dive: the three methods on one axis ---

Same question, three trade-offs.

- S_{GN}^* (norm): cheap, white-box, *magnitude* — “relevant?”
- S_{GI}^* (grad $\times$ input): cheap, white-box, *signed* — “towards target or foil?”
- S_E^* (erasure): expensive, black-box, *direct* — “measured, not approximated”.

Key insight

Gradients **approximate** a perturbation; erasure **performs** it. Erasure is the ground truth you cannot afford at scale.

🍷 Evaluation setup: BLiMP minimal pairs ---

- **BLiMP** — the Benchmark of Linguistic Minimal Pairs (Warstadt et al., 2020): 67 paradigms of minimal sentence pairs.
- They use **5 phenomena / 12 paradigms** (anaphor, argument structure, determiner-noun, NPI [negative-polarity items], subject-verb).
- Models: **GPT-2** (1.5B), **GPT-Neo** (2.7B).
- Each paradigm has a **rule** marking the token that enforces grammaticality.

Why BLiMP

They need an exam where the *correct* causal token is known *in advance* — so a method can be graded objectively.

🍷 Didactic aside: what is a “minimal pair”? ---

Two sentences that differ in **one** spot, one grammatical and one not:

“The **author laughs.**” (ok)

“The **author laugh.**” (*)

The single contrast isolates exactly one grammatical decision (subject–verb agreement).

Why it is perfect here

A minimal pair *is* a target/foil pair: *laughs* vs *laugh*. We already know the cause (the subject) — so we can check whether the method points there.

🍷 Deep dive: the alignment metrics ---

E = binary vector of “correct” evidence tokens; S = saliency scores.

- **Dot product** $S \cdot E$ — saliency on true evidence (higher = better).
- **Probes needed** — rank of the first true-evidence token (lower = better).
- **MRR** — mean reciprocal rank of that token (higher = better).

Remarks

All three ask: **does the method put the correct token near the top?**

Probes-needed and MRR are two views of the same ranking.

🍷 BLiMP: which input token is the cause? ---

Anaphor number agreement

Acceptable: “Many teenagers were helping **themselves**.”

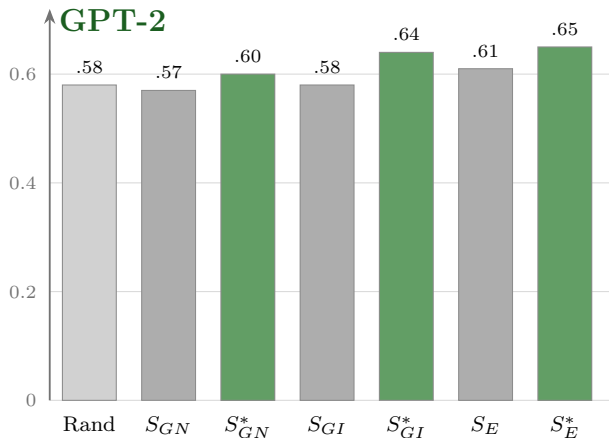
Unacceptable: “Many teenagers were helping **herself**.”

- The rule marks the antecedent “**teenagers**” as the evidence.
- A good explanation of *themselves vs herself* ranks **teenagers** high.

The test

Explaining *themselves vs herself*: if the method points at **teenagers** it is good; if it points elsewhere, it is poor.

🍷 Quantitative results: alignment on BLiMP ---



- Contrastive variants (S^* , green) align **better** with known evidence.
- The gain **grows with distance**: further evidence $\Rightarrow$ bigger contrastive advantage.

🍷 Human study: contrastive simulatability ---

Method	Acc. (%)
None	61.38
S_{GI}	64.00
S_{GI}^*	65.62
S_E	63.12
S_E^*	64.62

10 ML grad students predict GPT-2's output from input + explanation (4,000 judgments; model correct 50%).

Remarks

Contrastive explanations make the model more **predictable** to a human — that property is *simulatability*.

🍷 Use case: clustering decisions by their causes ---

- Represent each foil by its contrastive-saliency vector, then **k-means cluster**.
- Clusters recover grammatical categories **without supervision**:
 - ▶ target male pronoun → cluster of **female** pronouns;
 - ▶ animate noun → cluster of **inanimate** nouns;
 - ▶ singular noun → cluster of **plural** foils.

Key insight

These clusters differ from word-embedding neighbours — they reflect what the **model uses to decide**, not lexical similarity.

🍷 Limitations of Yin & Neubig ---

- **Foil selection is hard and consequential:** open-ended generation has no rule for which foil to contrast against — and changing the foil changes the result.
- **Faithfulness vs. plausibility:** matching human intuition does not *prove* the saliency captures the true computation.
- **Cost:** erasure is the most direct but scales poorly with length and foil-space size.

🍷 Concluding thoughts: connecting the two papers ---

Rajani et al. (2019)

- *Generated* natural-language explanations.
- Optimised for **plausibility** + accuracy.
- Cannot guarantee the explanation *caused* the prediction.

Yin & Neubig (2022)

- *Token-level, contrastive* saliency.
- Targets **faithful**, fine-grained evidence.
- Needs a foil; faithfulness still not proven.

The shared thread

Both attack *how do LMs reason* from opposite ends: **verbalise** the reasoning vs. **localise** the evidence.

🍎 Recap - the two papers at a glance ---

Recap. Both methods explain a trained **language model (LM)** *from the outside* (inputs $\rightarrow$ outputs) — but the LM itself differs per paper.

Paper 1 · Rajani et al. (2019) — CAGE

Verbalise the reasoning

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Paper 2 · Yin & Neubig (2022)

Localise the evidence

- **Explains:** a generative LM — GPT-2 / GPT-Neo (~ 50,000 tokens).
- **What:** attributes the output to input tokens.
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- **Family:** contrastive attribution (gradient/erasure).
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Open questions ---

- Can generated explanations (Paper 1) be **constrained to be faithful** via contrastive saliency (Paper 2)?
- What is the right way to **choose foils** for open-ended generation?
- For regulated settings (EU AI Act, Art. 13), is a **plausible** explanation enough, or is **faithfulness** legally required?

Discussion?

🍷 Thank you ---

Thank You!

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